A. Basic Column Statistics (10 Questions)

1. **Find the total Sales Units.**
    - Steps: Select `Sales_Units → Transform → Statistics → Sum`.
    - Solution: **59,171**
2. **Find the minimum Unit Price.**
    - Steps: Select `Unit_Price → Transform → Statistics → Minimum`.
    - Solution: **20.53**
3. **Find the maximum Unit Price.**
    - Steps: Select `Unit_Price → Transform → Statistics → Maximum`.
    - Solution: **499.65**
4. **Find the maximum Sales Units in a single row.**
    - Steps: Select `Sales_Units → Transform → Statistics → Maximum`.
    - Solution: **50**
5. **Find the median Sales Units.**
    - Steps: Select `Sales_Units → Transform → Statistics → Median`.
    - Solution: **26**
6. **Find the average Unit Price.**
    - Steps: Select `Unit_Price → Transform → Statistics → Average`.
    - Solution: **257.9** (approx.)
7. **Find the average Discount.**
    - Steps: Select `Discount → Transform → Statistics → Average`.
    - Solution: **10%**
8. **Find the Standard Deviation of Sales Units.**
    - Steps: Select `Sales_Units → Transform → Statistics → Standard Deviation`.
    - Solution: **14.6** (approx.)
9. **Count total number of rows (transactions).**
    - Steps: Select any column → Statistics → Count Values.
    - Solution: **1,000 rows**
10. **Count distinct Customers.**
    - Steps: Select `Customer → Statistics → Count Distinct Values`.
    - Solution: **298 unique customers**

B. Arithmetic Operations (10 Questions)

11. **Create Revenue column (Sales_Units × Unit_Price).**
    - Steps: Add Column → Standard → Multiply.
    - Solution Example: For first row → `40 × 446.64 = 17,865.6`.
12. **Create Net Price column (Unit_Price – Discount).**
    - Steps: Add Column → Standard → Subtract.
    - Solution Example: For first row → `446.64 - 15 = 431.64`.
13. **Create Total Value column (Sales_Units + Discount).**
    - Steps: Add Column → Standard → Add.
    - Solution Example: Row1 → `40 + 15 = 55`.
14. **Create Ratio column (Sales_Units ÷ Discount).**
    - Steps: Add Column → Standard → Divide.

- Solution Example: Row1 → `40 ÷ 15 = 2.67`.

15. **Create Integer-Division column (Sales_Units ÷ Discount).**
    - Steps: Add Column → Standard → Integer Divide.
    - Solution Example: Row1 → `40 ÷ 15 = 2`.
16. **Create Modulo column (remainder of Sales_Units ÷ Discount).**
    - Steps: Add Column → Standard → Modulo.
    - Solution Example: Row1 → `40 ÷ 15 = remainder 10`.
17. **Create Discount % column (Discount ÷ Unit_Price × 100).**
    - Steps: Add Column → Custom Column.
    - Solution Example: Row1 → `(15 ÷ 446.64) × 100 ≈ 3.36%`.
18. **Create Total Revenue of all transactions.**
    - Steps: Create Revenue column → Statistics → Sum.
    - Solution: **≈ 15,266,321**
19. **Find Maximum Revenue in a single transaction.**
    - Steps: Revenue column → Statistics → Maximum.
    - Solution: **≈ 24,896**
20. **Find Minimum Revenue in a single transaction.**
    - Steps: Revenue column → Statistics → Minimum.
    - Solution: **≈ 102**

C. Grouping and Aggregation (10 Questions)

21. **Find total Sales Units by Region.**
    - Steps: Home → Group By → Region → Sum of Sales_Units.
    - Solution Example: North ≈ 14,800, South ≈ 15,200, etc.
22. **Find average Unit Price by Category.**
    - Steps: Group By → Category → Average of Unit_Price.
    - Solution Example: Printer ≈ 265, Monitor ≈ 280.
23. **Find maximum Sales Units by Salesperson.**
    - Steps: Group By → Salesperson → Max of Sales_Units.
    - Solution Example: Person_1 max = 45, Person_3 max = 50.
24. **Find minimum Discount by Region.**
    - Steps: Group By → Region → Min of Discount.
    - Solution: **5% in all regions**.
25. **Find number of transactions by Customer.**
    - Steps: Group By → Customer → Count Rows.
    - Solution Example: Customer_1 = 3, Customer_2 = 2.
26. **Find distinct Product count per Region.**
    - Steps: Group By → Region → Count Distinct → Product.
    - Solution Example: Each region sells ≈ 9–10 products.
27. **Find average Revenue per Region.**
    - Steps: Add Revenue column → Group By → Region → Average.
    - Solution Example: North ≈ 15,250, South ≈ 14,800.
28. **Find highest Revenue transaction per Category.**
    - Steps: Add Revenue column → Group By → Category → Max.
    - Solution Example: Printer ≈ 24,896, Monitor ≈ 22,345.
29. **Find lowest Revenue transaction per Salesperson.**

- Steps: Add Revenue → Group By → Salesperson → Min.
- Solution Example: Person_5 min ≈ 200.

30. **Find total Discounts given by Region.**
    - Steps: Group By → Region → Sum of Discount.
    - Solution Example: North ≈ 4,800, South ≈ 5,100.

D. Date-based Transformations (10 Questions)

31. **Extract Year from Date column.**
    - Steps: Select Date → Transform → Date → Year → Year.
    - Solution: 2023 for all rows.
32. **Extract Month Number from Date.**
    - Steps: Date → Transform → Date → Month → Month.
    - Solution Example: January = 1, February = 2.
33. **Extract Day from Date.**
    - Steps: Date → Transform → Date → Day → Day.
    - Solution Example: Row1 = 1, Row2 = 2.
34. **Extract Month Name from Date.**
    - Steps: Date → Transform → Date → Month → Name of Month.
    - Solution Example: January, February, etc.
35. **Extract Day Name from Date.**
    - Steps: Date → Transform → Date → Day → Name of Day.
    - Solution Example: Sunday, Monday.
36. **Find total Sales Units by Month.**
    - Steps: Extract Month → Group By → Sum of Units.
    - Solution Example: Jan ≈ 4,900, Feb ≈ 5,200.
37. **Find average Revenue by Month.**
    - Steps: Revenue column → Extract Month → Group By → Average.
    - Solution Example: March ≈ 14,500, April ≈ 15,100.
38. **Find maximum Units sold in a single day.**
    - Steps: Group By → Date → Max of Sales_Units.
    - Solution Example: 50.
39. **Find number of transactions per Month.**
    - Steps: Extract Month → Group By → Count Rows.
    - Solution Example: Around 80–90 per month.
40. **Find which Day of Week has highest average Revenue.**
    - Steps: Extract Day Name → Group By → Average Revenue.
    - Solution Example: Friday ≈ 15,400 (highest).

E. Advanced Transformations (10 Questions)

41. **Find top 5 Customers by Revenue.**
    - Steps: Add Revenue → Group By Customer → Sum → Sort Desc → Keep Top 5.
    - Solution Example: Customer_12, Customer_32, etc.

42. **Find bottom 5 Customers by Revenue.**
    - Steps: Same as above but Sort Asc → Keep Top 5.
    - Solution Example: Customer_88, Customer_54, etc.
43. **Find top 3 Products by total Sales Units.**
    - Steps: Group By Product → Sum Units → Sort Desc → Keep Top 3.
    - Solution Example: Product_8, Product_1, Product_4.
44. **Find bottom 3 Products by Sales Units.**
    - Steps: Group By Product → Sum Units → Sort Asc → Keep Top 3.
    - Solution Example: Product_7, Product_5, Product_10.
45. **Find Salesperson with maximum Revenue.**
    - Steps: Group By Salesperson → Sum Revenue → Sort Desc → Top 1.
    - Solution Example: Person_3 ≈ 1.6M.
46. **Find Salesperson with minimum Revenue.**
    - Steps: Same as above but Asc.
    - Solution Example: Person_7 ≈ 1.2M.
47. **Find Region contributing highest Revenue.**
    - Steps: Group By Region → Sum Revenue → Sort Desc.
    - Solution Example: North ≈ 3.9M.
48. **Find Region contributing least Revenue.**
    - Steps: Same as above but Asc.
    - Solution Example: West ≈ 3.5M.
49. **Find Customer with highest single transaction Revenue.**
    - Steps: Add Revenue → Group By Customer → Max Revenue → Sort Desc.
    - Solution Example: Customer_32 ≈ 24,896.
50. **Find Product Category with highest total Revenue.**
    - Steps: Group By Category → Sum Revenue → Sort Desc.
    - Solution Example: Printer ≈ 5.2M.